SSL Certificate Watcher & AI Agent

Scan · Classify · Recommend · Monitor

UC-01 | Use Case & System Diagrams Report

Project	CertGuard AI	**Version**	v1.0	**Stack**	React · Python · FastAPI · Groq API · SQLite

Abstract

SSL Certificate Watcher & AI Agent is a proactive monitoring system designed to identify SSL certificates nearing expiration and help organizations avoid service disruptions. The application automatically scans domains, retrieves SSL certificate metadata, calculates remaining validity periods, classifies risk levels, and generates AI-powered remediation recommendations. The solution also provides dashboard analytics, historical tracking, and report-export capabilities.

1. Introduction

SSL certificates play a critical role in securing communication between users and web applications. Expired certificates result in browser warnings, service interruptions, and loss of customer trust. Manual monitoring of SSL certificates is time-consuming and prone to human error. This project automates certificate monitoring and provides intelligent recommendations through an AI-powered agent.

2. Problem Statement

Organizations often manage multiple websites and services, each protected by SSL certificates with different expiration dates. Key challenges include:

- Lack of centralised certificate monitoring.
- Manual tracking of expiry dates.
- Risk of unexpected certificate expiration.
- Service disruptions and security warnings.
- Increased operational effort.

An automated solution is required to continuously monitor SSL certificates and provide actionable insights.

3. Objectives

- Automatically scan SSL certificates of monitored domains.
- Retrieve certificate issuer and validity details.
- Calculate remaining days before expiration.
- Classify certificates based on urgency / risk level.
- Generate AI-powered renewal recommendations via Groq API (Llama 3).
- Store scan history in SQLite for future reference.
- Export reports in CSV and Markdown formats.
- Provide a user-friendly dashboard for real-time monitoring.

4. System Actors & Use Cases

4.1 Actors

Actor	Description
User / Admin	Adds domains, triggers scans, views dashboard, exports reports.
SSL Scanner	Establishes TLS handshake with target domains; extracts certificate metadata.
AI Agent (Groq/Llama 3)	Analyses risk status; generates ticket title, priority, and remediation steps.
SQLite Database	Persists domain list, scan results, risk classifications, and AI recommendations.
Email Alert System	Sends SMTP notifications for certificates in Critical or Warning state.
Export Engine	Produces CSV / Markdown artefacts on demand.

4.2 Use Case Summary

UC#	Use Case	Primary Actor	Brief Description
UC1	Manage Domains	User	Add, validate, and list domains for monitoring.
UC2	Scan SSL Certificate	SSL Scanner	Connect to port 443, perform handshake, extract metadata.
UC3	Classify Risk Level	System	Tag certificate as Critical (≤7d), Warning (≤30d), or Healthy (>30d).
UC4	Generate AI Recommendation	AI Agent	Produce ticket title, priority, description, business impact, and action.
UC5	View Dashboard	User	Display totals and doughnut/bar visualisations.
UC6	Track Scan History	SQLite	Persist each scan result for trend analysis.
UC7	Send Email Alert	Email System	Notify stakeholders when certificates enter Critical/Warning state.
UC8	Export Report	Export Engine	Generate CSV or Markdown report of current scan results.

5. Technology Stack

Layer	Technology
Frontend	React, TypeScript, HTML5, CSS3
Backend	Python, FastAPI, Socket Programming, SSL Libraries
Database	SQLite
AI / LLM	Groq API · Llama 3 (llama3-8b-8192)
Alert System	Gmail SMTP via smtplib
Reporting	CSV Export, Markdown Export, pandas

6. System Architecture

#	Layer	Key Components / Technology
1	UI Layer	React dashboard — domain input, risk badges, doughnut/bar charts, history table.
2	API Layer	FastAPI — /scan, /domains, /history, /export, /alerts endpoints.
3	SSL Scanner	Python ssl/socket/cryptography — port-443 handshake, metadata extraction.
4	Risk Classifier	Rule engine — maps days remaining to Critical / Warning / Healthy.
5	AI Agent	Groq API + Llama 3 — six-task autonomous agent generating remediation tickets.
6	Storage	SQLite — domains, scan results, risk status, AI recommendations, audit history.
7	Export Engine	pandas — CSV / Markdown report generation on demand.
8	Alert System	smtplib + Gmail SMTP — email notifications for Critical and Warning certs.

7. Functional Modules

7.1 Domain Management Module

Allows users to add new domains, manage the monitored list, validate domain inputs, and trigger certificate scans.

Inputs: Domain names entered by the user.

Outputs: Updated monitoring list persisted to SQLite.

7.2 SSL Certificate Scanner

Establishes a secure connection with target domains and retrieves certificate metadata including issuer, validity dates, and serial number.

Process: Connect on port 443 → SSL handshake → retrieve certificate → extract metadata → calculate expiry.

Outputs: Domain name, Certificate Issuer, Valid-From Date, Expiry Date, Serial Number.

7.3 Risk Classification Module

Classifies each certificate based on the number of days remaining before expiration.

7.4 AI Agent Module

Analyses certificate status using Groq API (Llama 3) and generates structured remediation tickets.

Generated Fields: Ticket title, Priority level, Description, Business impact, Recommended action.

Benefits: Reduces manual analysis, provides actionable guidance, improves operational efficiency.

7.5 Database Module

Uses SQLite to store domain information, scan results, risk status, AI recommendations, and historical records.

Benefits: Lightweight, fast, easy deployment, persistent storage.

7.6 Dashboard Analytics

Provides real-time overview with total domains monitored, critical/warning/healthy counts, and custom doughnut and bar visualisations.

7.7 Reporting Module

Supports CSV export for data analysis and Markdown export for technical documentation and team collaboration.

Days Remaining	Status	Action Required
≤ 7 Days	Critical	Immediate renewal — page on-call team.
≤ 30 Days	Warning	Schedule renewal — send alert notification.
> 30 Days	Healthy	No action required — continue monitoring.

8. Use Case Specifications

8.1 UC1 — Manage Domains

Primary Actor	User / Admin
Precondition	Application is running; SQLite database is initialised.
Main Flow	User enters domain name → system validates format → domain added to SQLite → scan initiated.
Exception	Invalid domain format → validation error shown; duplicate domain → warning displayed.
Postcondition	Domain persisted; certificate scan queued.

8.2 UC2 — Scan SSL Certificate

Primary Actor	SSL Scanner
Precondition	Valid domain in monitoring list; network reachable on port 443.
Main Flow	Connect to domain:443 → TLS handshake → extract cert → parse issuer/dates/serial → calculate days remaining.
Alt Flow	Connection refused / timeout → log error; mark domain as unreachable; retry on next cycle.
Postcondition	Scan result stored in SQLite; risk classifier invoked.

8.3 UC3 — Classify Risk Level

Primary Actor	System (Rule Engine)
Input	Days remaining from SSL Scanner.
Output	risk ∈ { Critical, Warning, Healthy }
Rule	Days ≤ 7 → Critical; Days ≤ 30 → Warning; Days > 30 → Healthy.
Postcondition	Risk tag stored; AI Agent triggered for Critical/Warning certs.

8.4 UC4 — Generate AI Recommendation

Primary Actor	AI Agent (Groq/Llama 3)
Trigger	Certificate classified as Critical or Warning.
Main Flow	Compose context prompt → POST to Groq API (llama3-8b-8192) → parse JSON response → store ticket.
Output Fields	Ticket title, priority level, description, business impact, recommended action.
Postcondition	Recommendation persisted; surfaced in dashboard and email alert.

8.5 UC5 — View Dashboard

Primary Actor	User / Admin
Main Flow	Open dashboard → system queries SQLite → render summary cards → render doughnut & bar charts.
Displayed Data	Total domains monitored, Critical count, Warning count, Healthy count, last-scan timestamps.
Postcondition	User has real-time visibility into certificate health.

8.6 UC6 — Track Scan History

Primary Actor	SQLite Database
Main Flow	After each scan, append row (domain, scan_time, expiry, days_remaining, risk, ai_ticket_id) to history table.
Postcondition	Historical data available for trend analysis and audit.

8.7 UC7 — Send Email Alert

Primary Actor	Email Alert System
Trigger	Certificate status transitions to Critical or Warning.
Main Flow	Build HTML email with cert details + AI recommendation → send via Gmail SMTP (smtplib) → log delivery.
Exception	SMTP failure → log error; retry up to 3 times; fallback to in-app notification.
Postcondition	Stakeholders notified; delivery recorded in SQLite.

8.8 UC8 — Export Report

Primary Actor	Export Engine (pandas)
Input	User selects export format: CSV or Markdown.
CSV Output	Columns: Domain, Issuer, Valid From, Expiry, Days Remaining, Risk, AI Recommendation.
Markdown Output	Structured report with table, risk summary, and AI ticket details for team documentation.
Postcondition	File available for download; export event logged.

9. System Workflow

Step	Action
Step 1	User enters a domain in the React dashboard.
Step 2	SSL Scanner connects to domain:443 and retrieves certificate details.
Step 3	Expiry information is extracted and days remaining calculated.
Step 4	Risk Classification module tags the certificate (Critical / Warning / Healthy).
Step 5	AI Agent (Groq/Llama 3) generates remediation recommendation for at-risk certs.
Step 6	Results are persisted to SQLite (domain, scan result, risk tag, AI ticket).
Step 7	Dashboard analytics are refreshed with updated counts and visualisations.
Step 8	Email Alert System notifies stakeholders if cert is Critical or Warning.
Step 9	User can export a CSV or Markdown report for operational use.

10. Key Features

Feature	Description
SSL Certificate Monitoring	Continuous automated scanning of all registered domains.
Expiry Detection	Precise calculation of days remaining for each certificate.
Risk Classification	Three-tier severity model: Critical, Warning, Healthy.
AI-Powered Recommendations	Groq + Llama 3 generate actionable remediation tickets.
Dashboard Analytics	Real-time summary cards with doughnut and bar charts.
Historical Tracking	Append-only scan history in SQLite for trend analysis.
Email Alert Notifications	Gmail SMTP alerts on Critical / Warning state changes.
CSV Export	Tabular data export for spreadsheet analysis.
Markdown Export	Formatted reports for technical documentation.
SQLite Storage	Lightweight, zero-config persistent data layer.

11. Risks & Mitigations

Risk	Mitigation
Groq API unavailability	Fallback to rule-based summary; retry with exponential backoff.

Risk	Mitigation
SMTP delivery failure	Retry up to 3 times; log failure; surface in-app notification.
Domain unreachable on :443	Log connection error; mark domain as unreachable; alert user.
SQLite data loss	Scheduled backup of DB file; export before destructive operations.
False risk classification	Hard rule overrides LLM output for ≤7 and ≤30 day thresholds.

12. Future Enhancements

- Slack and Microsoft Teams alert integrations.
- Automatic certificate renewal via ACME/Let's Encrypt.
- Multi-user authentication and role-based access control.
- Cloud deployment (AWS / GCP / Azure).
- Real-time continuous monitoring with configurable intervals.
- REST API for third-party integrations.

13. Success Metrics

Metric	Target
0 unexpected certificate expirations	All domains scanned at least once per day.
≥ 95% AI recommendation accuracy	Llama 3 tickets rated useful by admin.
100% alert delivery	Every Critical/Warning transition triggers email.
≤ 2 s scan time per domain	SSL handshake + metadata extraction within 2 seconds.
100% scan results in SQLite	No data loss between scan and persistence.

14. Conclusion

SSL Certificate Watcher & AI Agent provides an efficient and intelligent solution for monitoring SSL certificates and preventing service disruptions caused by certificate expiration. By combining automated scanning, risk classification, AI-generated remediation recommendations, dashboard analytics, email alerts, and export capabilities, the system enables organisations to manage certificate lifecycles proactively and improve operational reliability.